Algebra

CHAPTER 6

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CHAPTER 6

Inequalities Involving Means

DPP 6.1

Simple Mean of Two Positive Real Numbers

Single Correct Answer Type

1. Let $ab = 1$, then minimum value of $\frac{1}{a^4}+\frac{1}{4b^4}$ is

(a) 1 (b) 2
(c) 1/4 (d) 1/2

2. The minimum value of $\frac{x^4+y^4+z^2}{xyz}$ for positive real numbers x, y, z is

(a) $\sqrt{2}$ (b) $2\sqrt{2}$
(c) $4\sqrt{2}$ (d) $8\sqrt{2}$

3. If $x > 0$, then $\frac{x^n}{1+x+x^2+\ldots+x^{2n}}$

(a) $\le \frac{1}{2n+1}$ (b) $< \frac{2}{2n+1}$
(c) $\ge \frac{1}{2n+1}$ (d) $> \frac{2}{2n+1}$

4. If positive quantities a, b, c are in H.P, then which of the following is NOT true?

(a) $b < \frac{a+c}{2}$ (b) $\frac{1}{a-b}-\frac{1}{b-c}<0$
(c) $ac > b^2$
(d) $bc(1-a), ac(1-b), ab(1-c)$ are in H.P.

5. Let a, b, c and d be four positive numbers. Now, consider the following two statements:

S_1: $\left(\frac{a}{b}+\frac{c}{d}\right)\left(\frac{b}{c}+\frac{d}{e}\right) \ge 4\sqrt{\frac{a}{e}}$

S_2: $\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{e}+\frac{e}{a}$ cannot take value 4.

Then,

(a) Only S_1 is true
(b) Only S_2 is true
(c) Both S_1 and S_2 are false
(d) Both S_1 and S_2 are true

Numerical Value Type

6. Three positive numbers a, b and c are in A.P.. If $E = \frac{a+8b}{2b-a}+\frac{8b+c}{2b-c}$, then the least value of E is _____.

7. If a, b, c, d are four positive real numbers such that $abcd = 1$, then least value of $(a+4)(b+4)(c+4)(d+4)$ is _____.

8. If x, y, z are positive real numbers such that $x^2+y^2+z^2=7$ and $xy+yz+xz=4$, then the minimum value of $100xy$ is ____.

Answer Key

Single Correct Answer Type

1. (a) **2.** (b) **3.** (a) **4.** (d) **5.** (d)

Numerical Value Type

6. (18) **7.** (625) **8.** (25)

DPP 6.2

Simple Mean of *n* Positive Real Numbers

Single Correct Answer Type

1. If $x > 0, y > 0, z > 0$, then the least value of $x^{\log_e y - \log_e z} + y^{\log_e z - \log_e x} + z^{\log_e x - \log_e y}$ is

(a) 3 (b) 1
(c) 5 (d) 6

2. Let $27pqr \geq (p + q + r)^3$ and $3p + 4q + 5r = 12$, where $p, q, r \in R^+$. Then the value of $8p + 4q - 7r =$

(a) 2 (b) 3
(c) 4 (d) 5

3. If $n_1, n_2, n_3, \ldots, n_{100}$ are positive real numbers such that $n_1 + n_2 + n_3 + \ldots + n_{100} = 20$ and $k = n_1(n_2 + n_3 + n_4)(n_5 + n_6 + \ldots + n_9)(n_{10} + \ldots + n_{16}) \ldots (\ldots + n_{100})$, then k belongs to

(a) (0, 100] (b) (0, 128]
(c) [0, 144] (d) (0, 1024]

4. If a, b, c are the sides of a triangle, then the least value of $\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a}$ is

(a) 1/3 (b) 1
(c) 3 (d) 6

5. If x_1, x_2 and x_3 are the positive roots of the equation $x^3 - 6x^2 + 3px - 2p = 0, p \in R$, then the value of p is equal to

(a) 2 (b) 3
(c) 4 (d) 6

Numerical Value Type

6. The least integral value of

$$f(x) = \frac{(x-1)^7 + 3(x-1)^6 + (x-1)^5 + 1}{(x-1)^5} \ \forall\ x > 1 \text{ is } ____.$$

7. Given that x, y, z are positive real numbers such that $xyz = 32$. The minimum value of $\sqrt{(x+2y)^2 + 2z^2 - 15}$ is equal to ____.

8. If $x, y, z > 0$ and $x + y + z = 1$, then the least value of $\frac{2x}{1-x} + \frac{2y}{1-y} + \frac{2z}{1-z}$ is ____.

9. Let $x, y, z \in R^+$ and $2xy + 3yz + 4xz = 18$. If α, β and γ are the values of x, y and z, respectively, for which xyz attains its maximum value, then the value of $2\alpha + \beta + \gamma =$ ____.

Answer Key

Single Correct Answer Type

1. (a) **2.** (d) **3.** (d) **4.** (b) **5.** (c)

Numerical Value Type

6. (6) **7.** (9) **8.** (3) **9.** (6)

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DPP 6.3

Weighted Mean and Mean of mth Power

Single Correct Answer Type

1. The minimum value of

$$f(x) = \cos^2 x + \frac{\sec x}{4}, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ is}$$

(a) 3/2 (b) 3/4
(c) 3/8 (d) none of these

2. If x lies between –5 and 11, then the greatest value of $(11 - x)^3 (x + 5)^5$ is

(a) $6^5 \times 10^3$ (b) $6^3 \times 10^3$
(c) $6^3 \times 10^4$ (d) $6^3 \times 10^5$

3. If a, b, c are positive real numbers and $2a + b + 3c = 1$, then the maximum value of $a^4 b^2 c^2$ is equal to

(a) $\frac{1}{3 \times 4^8}$ (b) $\frac{1}{9 \times 4^7}$
(c) $\frac{1}{9 \times 4^8}$ (d) $\frac{1}{27 \times 4^8}$

Multiple Correct Answers Type

4. If a and b are positive numbers such that $a^2 + b^2 = 4$ and $(a^2 b)$ has maximum value, then

(a) $\frac{a}{b} = \sqrt{2}$ (b) $a = b$
(c) $(a^2 b)_{\text{max.}} = \frac{16}{3}$ (d) $(a^2 b)_{\text{max.}} = \frac{16}{3\sqrt{3}}$

5. If a and b are positive real numbers such that $a^2 b^3 = \frac{4}{9}$, then $(a + b)$ can be equal to

(a) $\frac{5}{3}$ (b) $\frac{5}{2}$
(c) 1.7 (d) 1.6

Linked Comprehension Type

For Questions 6–7

If a, b, x and y are real numbers such that $x, y > 0$, then $\frac{a^2}{x} + \frac{b^2}{y} \geq \frac{(a+b)^2}{x+y}$ as on solving it we have $(ay - bx)^2 \geq 0$.

Similarly, we can extend the inequality to three pairs of numbers i.e., $\frac{a^2}{x} + \frac{b^2}{y} + \frac{c^2}{z} \geq \frac{(a+b+c)^2}{x+y+z}$

Now, use this result to solve the following questions:

6. The value $\frac{a^2+b^2}{a+b} + \frac{b^2+c^2}{b+c} + \frac{a^2+c^2}{a+c}$

(a) $\geq (a + b + c)$ (b) $\geq \frac{1}{2}(a + b + c)$
(c) $\geq \frac{3}{2}(a + b + c)$ (d) None of these

7. If $abc = 1$, then the minimum value of

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(a+c)} + \frac{1}{c^3(a+b)} \text{ is}$$

(a) 3 (b) 3/2
(c) 6 (d) 9

Numerical Value Type

8. If $6x^2 + \frac{12}{x} > a \ \forall \ x > 0$, then the greatest integral value of a is _____.

9. Consider the three-dimensional figure represented by $xyz^2 = 2$, then its minimum distance from origin is _____.

10. If $a, b, c, d \in R^+$ and $a + b + c + d = 4$, then minimum value of $\left(a + \frac{1}{b}\right)^3 + \left(b + \frac{1}{c}\right)^3 + \left(c + \frac{1}{d}\right)^3 + \left(d + \frac{1}{a}\right)^3$ is _____.

Answer Key

Single Correct Answer Type

1. (b) **2.** (d) **3.** (c)

Multiple Correct Answers Type

4. (a, d) **5.** (a, b, c)

Linked Comprehension Type

6. (a) **7.** (b)

Numerical Value Type

8. (17) **9.** (2) **10.** (32)

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CHAPTER 6

DPP 6.1

Single Correct Answer Type

1. (a) Using A.M. ≥ G.M., we get

$$\frac{\frac{1}{a^4}+\frac{1}{4b^4}}{2} \geq \frac{1}{2} \qquad \text{(as } ab = 1\text{)}$$

$$\Rightarrow \quad \frac{1}{a^4}+\frac{1}{4b^4} \geq 1.$$

Hence, minimum value of $\left(\frac{1}{a^4}+\frac{1}{4b^4}\right)$ is 1.

2. (b) By A.M. ≥ G.M.

$x^4 + y^4 \geq 2x^2y^2$ and $2x^2y^2 + z^2 \geq \sqrt{8}\, xyz$.

$$\Rightarrow \quad \frac{x^4+y^4+z^4}{xyz} \geq \sqrt{8}$$

3. (a) $x+\frac{1}{x} \geq 2, \ldots, x^n+\frac{1}{x^n} \geq 2$

On adding all these inequalities, we get

$$\left(x+\frac{1}{x}\right)+\left(x^2+\frac{1}{x^2}\right)+\ldots+\left(x^n+\frac{1}{x^n}\right) \geq 2n,$$

$$\therefore \quad \left(\frac{1}{x^n}+\frac{1}{x^{n-1}}+\ldots\frac{1}{x}\right)+1+\left(x+x^2+\ldots+x^n\right) \geq 1+2n$$

$$\therefore \quad \frac{\left(1+x+\ldots+x^{n-1}+x^n\right)+x^{n+1}+x^{n+2}+\ldots+x^{2n}}{x^n} \geq 1+2n$$

$$\therefore \quad \frac{x^n}{1+x+\ldots+2^{2n}} \leq \frac{1}{1+2n}$$

4. (d) b is H.M. of a and c

$$b < \frac{a+c}{2} \text{ and } b < \sqrt{ac}$$

$$\Rightarrow \quad b^2 < ac$$

$$\Rightarrow \quad b-a < c-b$$

$$\Rightarrow \quad a-b > b-c$$

$$\Rightarrow \quad \frac{1}{a-b}-\frac{1}{b-c} < 0$$

also $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

So, $bc(1-a)$, $ac(1-b)$, $ab(1-c)$ are in A.P.

5. (d) $\frac{a}{b}+\frac{c}{d} \geq 2\sqrt{\frac{a}{b}.\frac{c}{d}}$ ...(i)

$\frac{b}{c}+\frac{d}{e} \geq 2\sqrt{\frac{b}{c}.\frac{d}{e}}$ ...(ii)

$$\therefore \quad \left(\frac{a}{b}+\frac{c}{d}\right)\left(\frac{b}{c}+\frac{d}{e}\right) \geq 4\sqrt{\frac{a}{e}}$$

Also, $\frac{\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{e}+\frac{e}{a}}{5} \geq \sqrt[5]{\frac{a}{b}.\frac{b}{c}.\frac{c}{d}.\frac{d}{e}.\frac{e}{a}} \geq 1$

$$\therefore \quad \frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{e}+\frac{e}{a} \geq 5$$

Numerical value type

6. (18)

$$E = \frac{a+4a+4c}{a+c-a}+\frac{4a+4c+c}{a+c-c} \qquad \text{(Since } a, b, c \text{ are in A.P.)}$$

$$= 5\left(\frac{a}{c}+\frac{c}{a}\right)+8$$

Since a and c are +ve, $\frac{a}{c}+\frac{c}{a} \geq 2$

$$\therefore \quad E \geq 10+8$$

$$\therefore \quad E \geq 18$$

7. (625)

Given $abcd = 1$

Using A.M. ≥ G.M., we get $a + b + c + d \geq 4$

For minimum value, $a = b = c = d = 1$

Therefore, minimum value of $(a + 4)(b + 4)(c + 4)(d + 4)$ is 625.

8. (25)

$xy = 4 - (x + y)z$

Now, consider $x + y$ and z.

Using G.M. ≥ A.M., we have

$$\sqrt{(x+y)z} \leq \frac{x+y+z}{2}$$

Also, $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx) = 7 + 8 = 15$

Now $(x+y)z \leq \frac{(x+y+z)^2}{4} \leq \frac{15}{4}$

$$\Rightarrow \quad (x+y)z|_{\max} = \frac{15}{4}$$

$$\Rightarrow \quad xy|_{\min} = 4-\frac{15}{4} = \frac{1}{4}$$

DPP 6.2

Single Correct Answer Type

1. (a) Let $\log_e x = a$, $\log_e y = b$, $\log_e z = c$.

$$\Rightarrow \quad x = e^a, y = e^b, z = e^c$$

So, given expression $e^{a(b-c)} + e^{b(c-a)} + e^{c(a-b)}$

Using A.M. ≥ G.M., we get

$$\frac{e^{a(b-c)}+e^{b(c-a)}+e^{c(a-b)}}{3} \geq \left[e^{a(b-c)+b(c-a)+c(a-b)}\right]^{1/3}$$

$$\therefore \quad e^{a(b-c)}+e^{b(c-a)}+e^{c(a-b)} \geq 3$$

2. (d) Given $27pqr \geq (p+q+r)^3$ ...(i)

Using A.M. ≥ G.M.

$\Rightarrow \quad \dfrac{p+q+r}{3} \geq (pqr)^{1/3}$ ...(ii)

$\Rightarrow \quad p = q = r$ (using (i) and (ii))

Also, $3p + 4q + 5r = 12$

$\Rightarrow \quad p = q = r = 1$

3. (d) Using A.M. ≥ G.M.

$$\frac{[n_1 + (n_2 + n_3 + n_4) + (n_5 + n_6 + \ldots + n_9) + \ldots + (\ldots + n_{100})]}{10} \geq \sqrt[10]{k}$$

$\Rightarrow \quad 2 \geq \sqrt[10]{k} \quad \Rightarrow \quad 0 < k \leq 1024$

4. (b) $c + a - b$, $b + c - a$, $a + b - c$ are all positive

Using A.M. ≥ G.M., we get

$$\therefore \quad \frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \quad \ldots(1)$$

$$\geq \left[\frac{abc}{(c+a-b)(a+b-c)(b+c-a)}\right]^{1/3}$$

Also, $a^2 \geq a^2 - (b-c)^2 \quad \Rightarrow \quad a^2 \geq (a+b-c)(a-b+c)$

Similarly, $b^2 \geq (b+c-a)(b-c+a)$,

and $c^2 \geq (c+a-b)(c-a+b)$

$\therefore \quad a^2b^2c^2 \geq (a+b-c)^2 (b+c-a)^2 (c+a-b)^2$

Thus, $abc \geq (a+b-c)(b+c-a)(c+a-b)$

$$\Rightarrow \quad \frac{abc}{(c+a-b)(a+b-c)(b+c-a)} \geq 1.$$

Hence, from (1) $\dfrac{abc}{(c+a-b)(a+b-c)(b+c-a)} \geq 1$

5. (c) $x^3 - 6x^2 + 3px - 2p = 0$

$$\text{A.M.} = \frac{x_1 + x_2 + x_3}{3} = \frac{6}{3} = 2$$

$$\text{H.M.} = \frac{3}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3}} = \frac{3x_1x_2x_3}{\Sigma x_1x_2} = 2$$

$\because \quad$ A.M. = H.M. $\quad \Rightarrow \quad x_1 = x_2 = x_3 = 2$

$\therefore \quad 2p = 2 \times 2 \times 2 \quad$ or $\quad p = 4$

Numerical Value Type

6. (6) $f(x) = (x-1)^2 + 3(x-1) + 1 + (x-1)^{-5}$

Consider $x - 1 = a > 0$ (as $x > 1$)

So, we have $a^2 + 3a + 1 + a^{-5}$

Consider quantities $a^2, a, a, a, 1, a^{-5}$.

Using A.M. ≥ G.M., we get

$$\frac{a^2 + a + a + a + 1 + a^{-5}}{6} \geq 1$$

$\Rightarrow \quad a^2 + 3a + 1 + a^{-5} \geq 6 \quad \Rightarrow \quad f(x) \geq 6 \ \forall\, x > 1.$

7. (9) $\dfrac{x^2 + 2xy + 2xy + 4y^2 + z^2 + z^2}{6} \geq \left(16x^4y^4z^4\right)^{1/6}$

$(x + 2y)^2 + 2z^2 \geq 96$

$(x + 2y)^2 + 2z^2 - 15 \geq 81$

Therefore, the least value of $\sqrt{(x+2y)^2 + 2z^2 - 15}$ is 9.

8. (3) $x, y, z > 0$

Using A.M. ≥ H.M., we get

$$\frac{(x+y)+(y+z)+(z+x)}{3} \geq \frac{3}{\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x}}$$

$$\Rightarrow \quad 2(x+y+z)\left(\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x}\right) \geq 9$$

$$\Rightarrow \quad 2\left(\frac{x+y+z}{x+y} + \frac{x+y+z}{y+z} + \frac{x+y+z}{z+x}\right) \geq 9$$

$$\Rightarrow \quad 2\left[\left(1 + \frac{z}{x+y}\right) + \left(1 + \frac{x}{y+z}\right) + \left(1 + \frac{y}{x+z}\right)\right] \geq 9$$

$$\Rightarrow \quad 2\left(\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}\right) \geq 3$$

$$\Rightarrow \quad 2\left(\frac{x}{1-x} + \frac{y}{1-y} + \frac{z}{1-z}\right) \geq 3 \qquad \{\because \ x + y + z = 1\}$$

9. (6) Using A.M. ≥ G.M., we get

$$\frac{2xy + 3yz + 4xz}{3} \geq \left(24x^2y^2z^2\right)^{\frac{1}{3}}$$

$\Rightarrow \quad 216 \geq 24x^2y^2z^2 \quad \Rightarrow \quad x^2y^2z^2 \leq 9 \quad \Rightarrow \quad xyz \leq 3$

So, xyz attains the greatest value when $2xy = 3yz = 4xz = 6$

$\Rightarrow \quad xy = 3, yz = 2, zx = 3/2$

$\Rightarrow \quad \alpha = 3/2, \beta = 2, \gamma = 1$

$\Rightarrow \quad 2\alpha + \beta + \gamma = 6$

DPP 6.3

Single Correct Answer Type

1. (b) $f(x) = \cos^2 x + \dfrac{\sec x}{4} = \cos^2 x + \dfrac{1}{4\cos x}$

For $x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$, $\cos x > 0$

Now, $f(x) = \cos^2 x + \dfrac{1}{8\cos x} + \dfrac{1}{8\cos x}$

Using A.M. ≥ G.M., we get

$$\Rightarrow \quad \frac{\cos^2 x + \frac{1}{8\cos x} + \frac{1}{8\cos x}}{3} \geq \left(\frac{1}{8^2}\right)^{\frac{1}{3}} = \left[\left(\frac{1}{4}\right)^3\right]^{\frac{1}{3}}$$

$$\Rightarrow \quad \cos^2 x + \frac{1}{4\cos x} \geq \frac{3}{4}$$

2. (d) Using weighted A.M. and G.M., we get

$$\frac{3\left(\frac{11-x}{3}\right) + 5\left(\frac{x+5}{5}\right)}{3+5} \geq \left(\left(\frac{11-x}{3}\right)^3 \left(\frac{x+5}{5}\right)^5\right)^{\frac{1}{3+5}}$$

$$\therefore \quad \frac{11-x+x+5}{8} \geq \left(\frac{(11-x)^3(x+5)^5}{3^3.5^5}\right)^{1/8}$$

$$\therefore \quad (2)^8 \geq \frac{(11-x)^3(x+5)^5}{3^3 \cdot 5^5}$$

$$\therefore \quad (11-x)^3 (x+5)^5 \leq 2^8 \cdot 3^3 \cdot 5^5$$

$$\Rightarrow \quad (11-x)^3 (x+5)^5 \leq 6^3 \cdot 10^5$$

3. (c) Consider positive numbers,

$$\frac{2a}{4}, \frac{2a}{4}, \frac{2a}{4}, \frac{2a}{4}, \frac{b}{2}, \frac{b}{2}, \frac{3c}{2}, \frac{3c}{2}$$

Using A.M. ≥ G.M., we get

$$\frac{\frac{2a}{4}+\frac{2a}{4}+\frac{2a}{4}+\frac{2a}{4}+\frac{b}{2}+\frac{b}{2}+\frac{3c}{2}+\frac{3c}{2}}{8} \geq \left(\frac{\frac{2a}{4}\cdot\frac{2a}{4}\cdot\frac{2a}{4}\cdot\frac{2a}{4}\cdot\frac{b}{2}\cdot\frac{b}{2}\cdot\frac{3c}{2}\cdot\frac{3c}{2}}{8}\right)^{\frac{1}{8}}$$

$$\Rightarrow \quad \frac{2a+b+3c}{8} \geq \left(\frac{3^2}{2^8}\cdot a^4b^2c^2\right)^{\frac{1}{8}}$$

$$\Rightarrow \quad \frac{1}{8} \geq \frac{\left(3^2\, a^4b^2c^2\right)^{\frac{1}{8}}}{2} \quad \Rightarrow \quad a^4b^2c^2 \leq \frac{1}{4^8 \times 3^2}$$

Multiple Correct Answers Type

4. (a, d)

Consider quantities $a^2/2$, $a^2/2$ and b^2.

Using A.M. ≥ G.M., we get

$$\frac{\frac{a^2}{2}+\frac{a^2}{2}+b^2}{3} \geq \left(\frac{a^2}{2}\cdot\frac{a^2}{2}\cdot b^2\right)^{\frac{1}{3}}$$

$$\Rightarrow \quad \left(\frac{4}{3}\right)^{\frac{3}{2}} \geq \left(\frac{(a^2b)^{\frac{2}{3}}}{2^{\frac{2}{3}}}\right)^{\frac{3}{2}}$$

$$\Rightarrow \quad (a^2\, b) \leq \left(\frac{4}{3}\right)^{\frac{3}{2}} \times 2 = \frac{16}{3\sqrt{3}}$$

$$\Rightarrow \quad (a^2\, b)_{\max} = \frac{16}{3\sqrt{3}} \text{ and } \frac{a^2}{2} = b^2 \quad \Rightarrow \quad \frac{a}{b} = \sqrt{2}$$

5. (a, b, c)

$$\frac{2\left(\frac{a}{2}\right)+3\left(\frac{b}{3}\right)}{5} \geq \left(\frac{a^2.b^3}{2^2\ 3^3}\right)^{\frac{1}{5}}$$

$$\Rightarrow \quad a+b \geq 5\left(\frac{4}{9(4)(27)}\right)^{\frac{1}{5}}$$

$$\Rightarrow \quad (a+b) \geq \frac{5}{3}$$

Linked Comprehension Type

6. (a) $\dfrac{a^2+b^2}{a+b}+\dfrac{b^2+c^2}{b+c}+\dfrac{a^2+c^2}{a+c}$

$$= \frac{a^2}{a+b}+\frac{b^2}{b+c}+\frac{c^2}{a+c}+\frac{b^2}{a+b}+\frac{c^2}{b+c}+\frac{a^2}{a+c}$$

$$\geq \frac{(2a+2b+2c)^2}{4(a+b+c)} \geq (a+b+c)$$

$$\Rightarrow \quad \frac{a^2}{a+b}+\frac{b^2}{b+c}+\frac{c^2}{a+c}+\frac{b^2}{a+b}+\frac{c^2}{b+c}+\frac{a^2}{a+c} \geq a+b+c$$

7. (b) $\dfrac{1}{a^3(b+c)}+\dfrac{1}{b^3(a+c)}+\dfrac{1}{c^3(a+b)}$

$$= \frac{1/a^2}{ab+ac}+\frac{1/b^2}{ab+bc}+\frac{1/c^2}{ac+bc} \geq \frac{\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)^2}{2(ab+bc+ac)} \quad \text{...(i)}$$

Now, $\left(\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c}\right)^2 = \dfrac{(ab+bc+ac)^2}{(abc)^2}$

$$= (ab+bc+ac)^2 \qquad (\text{As } abc = 1)$$

From (i), $\dfrac{1}{a^3(b+c)}+\dfrac{1}{b^3(a+c)}+\dfrac{1}{c^3(a+b)} \geq \dfrac{(ab+bc+ac)}{2}$

$$\geq \frac{3\,.\sqrt[3]{(abc)^2}}{2} \geq \frac{3}{2} \quad (\text{Using A.M.} \geq \text{G.M.})$$

Numerical Value Type

8. (17) $x > 0$

$$\therefore \quad 6x^2+\frac{12}{x} = 6x^2+\frac{6}{x}+\frac{6}{x} \geq 3(6^3)^{1/3}$$

$$\Rightarrow \quad 6x^2+\frac{12}{x} \geq 18 \quad \Rightarrow \quad a < 18$$

9. (2) Distance $= \sqrt{x^2+y^2+z^2}$

$$\therefore \quad x^2+y^2+\frac{2}{xy} = x^2+y^2+\frac{1}{xy}+\frac{1}{xy} \geq 4$$

$\therefore$ Minimum distance = 2

10. (32) Using A.M. ≥ H.M.

$$\frac{a+b+c+d}{4} \geq \frac{4}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}$$

Also, $\dfrac{\left(a+\frac{1}{b}\right)^3+\left(b+\frac{1}{c}\right)^3+\left(c+\frac{1}{d}\right)^3+\left(d+\frac{1}{e}\right)^3}{4}$

$$\geq \left(\frac{a+\frac{1}{a}+b+\frac{1}{b}+c+\frac{1}{c}+d+\frac{1}{d}}{4}\right)^3$$

$$\Rightarrow \quad \left(a+\frac{1}{b}\right)^3+\left(b+\frac{1}{c}\right)^3+\left(c+\frac{1}{d}\right)^3+\left(d+\frac{1}{e}\right)^3 \geq 4\left(\frac{4+4}{4}\right)^3 = 32$$